

COMPUTER NETWORKS

Week-8

4. Transport Layer

- Break big data into smaller pieces
- Large data is divided into segments
- Each segment is sent separately
- At the receiver → segments are reassembled (desegmentation)
- Sending a large video → divided into small packets → reassembled at receiver

Transport Layer

- This layer provides transparent transfer of data between end users
- Also controls the reliability of a given link through flow control, segmentation/desegmentation, and error control

Transport Layer

- Ensures that the data units are delivered error free
- Ensures that data units are delivered in sequence
- Ensures that there is no loss or duplication of data units
- Provides connectionless or connection oriented service
- Provides for the connection management

Transport Layer

- The transport layer header contains segment number (sequencing)
- Responsible for connection between two end ports (establish, transfer, release)

Transport Layer

- Flow Control
- Controls the speed of the data transmission
- Prevents overwhelming at the receiver end
- A slow mobile phone receiving data from a fast server → Transport layer slows down the sender

Transport Layer

- Error Control
- Detects lost or corrupted data
- Requests retransmission
- Resend request for the lost data packet or all data..

Transport Layer

- Connection Control
- Connection-oriented (reliable)
- Connectionless (fast but less reliable)
- TCP → reliable (used in web, email)
- UDP → faster (used in video streaming, gaming)
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TCP- Reliable Communication

- **Connection Setup (3-way handshake)**
- PC A → Server B: “*Can we communicate?*” (SYN)
- Server B → PC A: “*Yes, ready!*” (SYN-ACK)
- PC A → Server B: “*Let’s start!*” (ACK)
- **Data Transfer**
- Data is broken into **segments**
- Each segment has a **sequence number**
- **Acknowledgment**
- Server sends **ACK** after receiving each segment
- e.g.:
- Segment 1 received → ACK sent
- Segment 2 lost → no ACK sent

TCP- Reliable Communication

- **Retransmission**
- Sender detects missing ACK → **resends segment 2**
- **Order Guarantee**
- Even if segments arrive out of order, TCP rearranges them
- **Reliable delivery**
- Extra overhead (ACKs, retransmissions)
- Slower but accurate
- e.g. Opening a website (HTTP/HTTPS), Downloading a file from a server

UDP : User Datagram Protocol

- Fast Communication

UDP

- **No connection setup**
- PC A just starts sending data
- **Data Transmission**
- Data sent as **datagrams**
- No sequence tracking (or minimal)
- **No Acknowledgment**
- Server does NOT send ACKs
- **No Retransmission**
- If a packet is lost → ignored

UDP

- Adv:
- Faster
- No delay due to retransmission
- Possible data loss
- e.g. Live video lecture (Zoom, MS-Teams), Online gaming (real-time actions)

Example

- Watching a YouTube Video
- If using **TCP**
- Every frame must arrive correctly
- If one packet is missing → resend → causes buffering 😞
- If using **UDP**
- Missing packets are ignored
- Video continues smoothly
- Maybe a slight glitch, but no pause
- That's why streaming prefers **UDP**

Pont to note..

- TCP controls the network for reliability
UDP trusts the network for speed

Transport vs Network Layer

- Network Layer → Gets the data to the correct network
- Transport Layer → delivers it to the destination end system
- The Transport Layer ensures reliable, ordered, and correct delivery of data between applications on different devices